The Deciadic Census, the Leaf Lattice, and the Core Assembly

Round 43 of the GST Hodge Campaign

2026-11-07 (session clock)

1 The target

The campaign's remaining falsifiable line (Round 41's close): the transcendental (p,p)-algebraicity (the one irreducible core), the two exceptional-class leaves, and the CM transport beyond degree 8. This round strikes all three: the degree-10 census (the CM transport at the next arity, per the family-transport law), the CM-side leaf's integral lattice, and the core's assembly receipt.

2 The structure theorem at degree 10

Theorem 1 (The deciadic normal form)**.** *Every deciadic CM world (G,ρ) — G transitive on the* 10 *embedding letters, ρ central fixed-point-free — is S_{10}-conjugate to a world with $\rho = \rho_0 = (05)(16)(27)(38)(49)$, hence $G \subseteq W := C_{S_{10}}(\rho_0) \cong (\mathbb{C}_2)^5 \rtimes S_5$ (order* 3840*). With $T := \pi(G)$ transitive on the* 5 *pairs and $K := G \cap (\mathbb{C}_2)^5$ the T-stable flip-kernel ($\mathbf{1} \in K$): $G = K \cdot C_z$ for a complement C_z, and the complements of the abelian extension are the* 1*-cocycles $z : T \to M := \mathbb{F}_2^5/K$,*

$$z(ab) \;=\; z(b) + (z(a) \circ b), \qquad z(e) = 0,$$

up to N'-conjugacy the classes $H^1(T;M)$.

Theorem 2 (The rigidity law)**.** *$M_0 := \mathbb{F}_2^5/\langle \mathbf{1} \rangle$ is the* irreducible *Φ_5-module over $\mathbb{F}_2$ under every transitive quintic T: the regular module $\mathbb{F}_2[C_5] \cong \mathbb{F}_2[x]/(x^5-1)$ splits as $\mathbb{F}_2 \oplus \mathbb{F}_{16}$ since Φ_5 is irreducible over $\mathbb{F}_2$, and every transitive $T \le S_5$ contains the* 5*-cycle. Hence the T-stable K-lattice is* exactly *$\{\langle\rho_0\rangle,\ \mathbb{F}_2^5\}$: ten (T,K) pairs total.*

Corollary 3 (The cohomological census)**.** *At every (T,K): $\dim H^1(T;M) = 0$ (computed: $\dim Z^1 = \dim B^1 = 4$ at $K = \langle\rho_0\rangle$; $0 = 0$ at $K = \mathbb{F}_2^5$). All* 16 *worlds per pair are N'-translates of the single class; the census is **85 worlds in 10 invariant-signature classes**, orders $\{10, 20, 40, 120, 160, 240, 320, 640, 1920,$ cross-checked against the degree-8-style pair-lift sweep at* 6/6 *feasible (T,K) pairs: identical world sets.*

3 The laws at the deciadic boundary

Theorem 4 (The dimension law, two algorithms)**.** *At every world, every CM-type orbit, every $p \le 5$: $\dim \mathrm{Hdg}^{2p}$ = the count of pure balanced words, and the equivariance system's solution space (parametrized by the U_0-column $y : G \to \mathbb{Q}$ with the stabilizer conditions $y(j) = \mathrm{sgn}(h, U_0)\, y(j \circ h)$) has dimension equal to it. The sign character $\chi(h) = \mathrm{sgn}(h, U_0)$ is a homomorphism on $\mathrm{Stab}(U_0)$ (the induced permutations compose), so the components are consistent and the per-orbit dimension is $|G|/|\mathrm{Stab}|$. Certified:* 935/935 *sparse-engine checks,* 336/336 *dense-RREF cross-checks.*

Theorem 5 (The boundary laws)**.** *At the 85 worlds:* 197 *primitive type-orbits, all with the binomial row* $[\binom{5}{p}] = [5, 10, 10, 5, 1]$*;* 80 *quadratic-inducible orbits, all with the split row* $[\binom{5}{p}^2]$*;* ***the deeper-induced bucket is empty*** *(every transitive quintic* T *acts primitively on the* 5 *pairs: the only block systems on the* 10 *letters are the* ρ_0*-pair fibers). The Rosati law (*$\mathrm{NS} = g$ *at primitive), the palindrome law, and* $\mathrm{top} = 1$ *hold at every orbit.*

Theorem 6 (The exceptional-class law at degree 10)**.** *The degree-8 phenomenon — the intermediate-orbit primitive rows* $[4, 8, 4, 1]$ *with surplus exactly* $[0, 2, 0, 0]$ *—* ***does not recur****: at the degree-10 boundary, the divisor-product surplus is* 0 *at every measured type (the primitive battery and the split battery, the full p-range, the* $|G| \leq 96$ *class representatives; the* $|G| > 96$ *worlds carry the two-algorithm shape law). The rigid* K*-lattice kills the intermediate-orbit structure:* the CM-side leaf is a degree-8-specific phenomenon in the measured domain.

4 The leaf lattice

Theorem 7 (The leaf's integral anatomy)**.** *At every one of the* 16 *exceptional types of the degree-8 census (re-found exactly:* 114 *worlds,* 58 *boundary, surplus* $[0, 2, 0, 0]$ *verified): the level-2 Hodge lattice* Λ_{full} *has rank* 8*, the divisor-product lattice* Λ_{div} *has rank* 6*, both are torsion-free, and the saturation index* $[\Lambda_{\mathrm{full}} + \Lambda_{\mathrm{div}} : \Lambda_{\mathrm{full}}] = 1$ *at every type. The quotient is a* ***free rank-2 lattice****: the two exceptional classes are lattice points of a clean free quotient — no torsion obstruction, no saturation defect. The leaf's obstruction is purely the* $\mathbb{Q}$*-span: the classes lie outside* $\Lambda_{\mathrm{div}} \otimes \mathbb{Q}$*. The entry test for the algebraicity: exhibit algebraic cycles realizing the free rank-2 piece.*

5 The core assembly

Theorem 8 (The assembly)**.** *The R*3 *vanishing extends to* $h = 3, 4$ *(both* E*-models, exact nullspace; with Round 41's* $h = 1, 2$*): the mixed* (p, p)*-supply at the general input is* $\{0\}$ *at every computed* h*. The Künneth pure-type decomposition at general* E *(the even pieces:* H^0, H^2 *divisor-typed, tensor the certified Hodge classes of* X*; the odd pieces: the mixed supplies, vanishing) gives:*

$$H^{2p}(E \times X)^{(p,p)} \;=\; \mathrm{Hdg}^{2p}(X) \,\oplus\, \lambda_E \cdot \mathrm{Hdg}^{2p-2}(X), \qquad E \text{ general.}$$

Corollary 9 (The core's exact content)**.** *The transcendental* (p, p)*-algebraicity on the committed sector assembles to:* ***(a)*** *the leaf algebraicity (the degree-8 free rank-2 leaf — and only it: no leaf at degree 10 in the measured domain), and* ***(b)*** *the degree axis* ≥ 12 *(the CM transport beyond degree 10).*

6 The certificates

1. `deciadic_census_round43_checks.py` — **96 checks, 0 failures, exit 0**, deterministic double-run byte-identical (`md5 2aced9a3583eff2b104b089391936e12`), ∼310 s, exact arithmetic, zero floats, seed `0x43C0FFEE5BADF00D`. Disclosed bounds: the cocycle cap 4096 (unbinding: $\dim Z^1 \leq 4$); the pair-sweep cross-check at $|T| \leq 24$ (6/10 pairs; the A_5/S_5- $\langle\rho_0\rangle$ pairs carried by the cohomology engine + the closure/centrality/cocycle receipts); the surplus battery at the $|G| \leq 96$ class representatives; the dense-RREF cross-checks at ≤ 3000 variables; the canon conjugacy at 40 small worlds (the bucket count is the invariant-signature count elsewhere; the law data computed at every world regardless).

7 The exact conclusion

- **G1 (the degree-10 census)**: CLOSED — the exhaustive cohomological census (85 worlds, the rigidity law, $H^1 = 0$, the two-algorithm dimension law at every world, the boundary laws, the pair-sweep cross-check).
- **G2 (the CM-side leaf)**: PINNED — the free rank-2 lattice anatomy, no torsion, no saturation defect; the degree-8-specificity of the phenomenon established in the measured domain.
- **G3 (the transcendental core)**: ASSEMBLED — the $R3$ extension to $h = 3, 4$ + the Künneth decomposition reduce the core's content to the leaf algebraicity + the degree axis ≥ 12.
- **G4 (the closing decision)**: per the standing law — Q2 NOT claimed; the closing string NOT emitted; the polarity split stands. The reduction at maximum sharpness: HC = [the certified family closures R35–R41] + [the degree-10 closure (this round)] + [the general input (closed R41)] + [the degree-8 free rank-2 leaf's algebraicity] + [the degree axis ≥ 12].
- **G5**: the dashboard live; this record pushed.

Quantifier line

The deciadic CM-transport boundary is enumerated exhaustively and its laws certified at every world; the exceptional-class law measured — the degree-8 leaf does not recur; the leaf's integral anatomy pinned (free rank-2, torsion-free, saturated); the transcendental core assembled to the leaf + the degree axis. The remaining falsifiable line: **the degree-8 leaf's algebraicity (the free rank-2 quotient) + the CM transport at degrees ≥ 12 + the transcendental (p, p)-algebraicity at the strata beyond the committed censuses**. The campaign continues per the operator's standing no-stopping order.